

High-pressure processing of foods: a narrative review of when a named cycle is not one outcome

Abstract

Heat remains the default kill step in high-moisture foods because a named time–temperature cycle can be verified, written into law, and compared across plants. Hydrostatic high-pressure processing offers an alternative when flavour, colour, or native proteins are the reason heat is refused, typically as several minutes at a few hundred megapascals in a chilled or ambient vessel. Vegetative bacteria and yeasts in high-moisture packs often lose culturability under those holds, yet bacterial spores, several oxidases, and some legal pasteurisation bars do not move with plate counts. The literature that can be compared is mixed in kind: one scientific opinion on legal performance criteria, one mixed-model systematic review of bovine milk, laboratory and semi-industrial challenges in juices, dairy, meats, and seafood, and narrative reviews of kinetics, proteins, aroma, and phenolics. The hinge is simple. A labelled megapascal number is a process setting, not a pasteurisation claim, because ice, water activity, organic acids, strain, recovery agar, and adiabatic heat change the log reduction that the same hold appears to deliver. Those patterns do not imply that chilled commercial pressure is interchangeable with high-pressure homogenization, with heat-assisted pressure, or with a five-log juice or milk criterion. Process specifications therefore have to name matrix, assay, and whether heat was part of the kill, rather than copy a pressure from another food.

Keywords

high-pressure processing; high-pressure homogenization; bacterial spores; milk pasteurisation; sublethal injury; polyphenol oxidase

Key Summary Points

- Industrial ambient high-pressure processing commonly uses 400–600 MPa for a few minutes and inactivates many vegetative cells in high-moisture foods; the evidence grade is a scientific opinion plus multiple storage trials, not a pooled log reduction.
- The same megapascal number is not one kill: freezing, water activity, lactate, strain, agar, and hold versus pulse change measured log reductions in primary challenges.

- Legal five-log juice language and legal thermal pasteurisation of ruminant milk are performance criteria. Included juice trials often measured indigenous flora, and the milk systematic review did not meta-analyse a paired 72 °C/15 s contrast.
- Ambient pressure does not inactivate bacterial spores at commercial settings; heat-assisted pressure is a different process because adiabatic compression heating is part of the kill.
- Quality endpoints (oxidases, colour, texture, whey proteins, allergenicity, aroma, phenolics) can improve or worsen independently of plate counts.
- High-pressure homogenization is a shear machine. A cooled-valve apple-juice trial at 100–200 MPa reached at most 1.4 log of native aerobes, which is not a hydrostatic pasteurisation equivalent.

1. Introduction

Heat is the default kill step in food preservation because a named time–temperature cycle can be specified, verified, and written into law. Legal pasteurisation of ruminant milk is still defined as at least 72 °C for 15 s or 63 °C for 30 min, with alkaline phosphatase used as a process indicator, and those cycles are expected to deliver more than 10 log₁₀ of several vegetative pathogens [1]. Juice hazard-analysis rules that ask for a five-log reduction of a pertinent pathogen are likewise performance criteria: they name an outcome, not a machine. When a processor wants to keep the colour of a puree, the whey proteins of milk, or the raw appearance of a seafood fillet, heat is the thing being avoided, and some other unit operation has to carry the microbial bar. That substitution is not automatic. A process that looks cold at the vessel wall can still heat the pack from inside, and a process that looks lethal on agar can leave spores, enzymes, or injured cells behind.

High-pressure processing (HPP) is an isostatic hold in which a packed food sits in a liquid, usually water, and the liquid transmits pressure uniformly so that piece size does not create a cold spot the way a retort can [1,2]. Industrial cycles for chilled high-moisture foods most often use 400–600 MPa for about 1.5–6 min, often with 4–8 °C transmitting water and with product temperature kept below 45 °C so that the treatment stays in the non-thermal class used by the European Food Safety Authority (EFSA) Biological Hazards Panel [1]. Vegetative spoilage organisms and many Gram-negative pathogens in juices and purees are generally inactivated in that window, and reviews of plant-food kinetics treat 400–600 MPa as the commercial vegetative range while stating that pasteurisation language still requires at least 5–6 log colony-forming units per gram or millilitre [2,3]. Bacterial endospores do not follow that pattern. Ambient industrial pressures do not inactivate spores, which is why low-acid shelf-stable claims cannot be copied from a chilled juice cycle [1,3]. Low-moisture foods with

less than about 40% water are usually not treated, because inactivation is low when there is too little free water to transmit the mechanical load [1].

Compression is not thermodynamically cold. Work of compression raises product temperature, ideally uniformly, by a rule of thumb of about 3 °C per 100 MPa in water-rich foods from room temperature, so that a 600 MPa hold can add about 18 °C before the vessel begins to lose heat to the wall [4]. Fats heat more than water. When that adiabatic rise is allowed, or when the incoming product is already warm, the process is high-pressure thermal processing (HPTP), also discussed as pressure-assisted thermal sterilization, and heat is then part of the kill rather than an accident of the pump [3,4]. The same megapascal number at chill temperature and at 70–85 °C is therefore not one technology. High-pressure homogenization (HPH) is a third machine: fluid is forced through a narrow valve or orifice, typically at tens to a few hundred megapascals, and shear, cavitation, and impact do the work that hydrostatic compression does not [5]. Homogenization cannot be run on a solid ham or a vacuum-packed fillet, and a 200 MPa homogenizer label is not a 200 MPa isostatic hold.

The biology that those machines meet is not uniform either. Vegetative membranes compress, proteins lose non-covalent assemblies, and nucleic acids can leak; those events are consistent with plate-count death when the cell cannot repair [3,6]. Pores that take up propidium iodide can shrink during chilled storage, so a large immediate log reduction is not proof that every treated cell is dead [6]. Culture can also miss cells that enter a viable but nonculturable (VBNC) state and later resuscitate, which is why a blank plate after pressure is an assay result rather than a sterility certificate [6]. Gram-positive organisms, including *Listeria monocytogenes* and *Staphylococcus aureus*, are often more barotolerant than Gram-negatives, and stationary-phase cells are more tolerant than exponential-phase cells, but matrix protection can reverse those orders [3]. Enzymes that cause browning, cloud loss, or texture change unfold on a different curve from membranes, so polyphenol oxidase (PPO) can remain active after a cycle that has already dropped aerobic plate counts [2].

Legal language sits on top of that physics and does not dissolve it. A five-log juice claim and a milk-pasteurisation claim are bars that a named cycle may or may not meet, depending on organism, fat, pH, and hold [1,7]. Mixed-model averages from bovine-milk trials put the pressure needed for a greater-than-five-log vegetative reduction near 450 MPa for *L. monocytogenes*, 500 MPa for *Salmonella enterica*, and 600 MPa for *S. aureus*, while several *Escherichia coli* strains still miss that bar at 600 MPa when results are averaged [7]. Those averages are not a plant specification, because come-up rate, fat, and strain scatter the same pressure across more than one log [7]. Combination treatments with ultrasound, organic acids, bacteriocins, or mild heat are often proposed when single room-temperature pressure is

limited for Gram-positives and spores [8]. They are additional variables, not proof that the neighbouring technology is interchangeable with the hydrostatic hold.

The live controversy is therefore not whether pressure can reduce vegetative counts in high-moisture foods. It can, under the conditions studied. The controversy is when a labelled megapascal number is treated as a pasteurisation equivalent, a spore kill, or a quality guarantee that travels from juice to milk to ham. Processors already sell chilled high-pressure juices, guacamole, ready-to-eat (RTE) meats, and seafood; those products succeed as refrigerated foods. They do not, on the included evidence, succeed as legal thermal pasteurisation of ruminant milk, as a generic five-log pathogen validation at every juice pressure, or as ambient spore sterilization.

The central argument of this review is that a named hydrostatic cycle is a setting whose microbial and quality meaning is fixed by matrix, ice, solute, injury assay, hold schedule, and whether heat was allowed to rise, so that commercial chilled success and a legal five-log or milk-pasteurisation claim remain different questions.

2. Methods

2.1 Search and sources

Records were identified from a Scopus bibliographic export of journal articles and reviews on nonthermal food processing and high pressure, dated 20 September 2026 (359 records). Titles and metadata were screened first. Full texts were opened only when a public publisher PDF or public Europe PMC JATS file was available. Paywalled files were not opened. Related records were sought in OpenAlex for 2021–2026 peer-reviewed journals. After the literature table was built, five targeted OpenAlex searches were run for interpretation gaps (paired milk high-pressure versus high-temperature short-time trials, industrial juice five-log pathogen validations, *Clostridium botulinum* or proteolytic spore data, primary homogenization inactivation, and freeze-then-pressure *Listeria* trials). Public files that filled those gaps were added; files without a public PDF were left unread.

2.2 Eligibility

Eligible papers named high pressure, HPP, high hydrostatic pressure, HPH, or a listed neighbouring nonthermal operation (ultrasound, pulsed electric fields, cold plasma, irradiation, ohmic or radio-frequency heating, hydrodynamic cavitation) in a human-food, juice, meat, dairy, produce, or microbial-inactivation context, for 2021–2026 journal articles and reviews. Bibliographic export records in that window stayed eligible even if older papers exist in the field. Books, book chapters, conference papers, retracted items, pet-food papers, and titles without a named technology or food-processing context were excluded. HPP is the

organising process; other unit operations appear only where they were measured beside pressure. Reviews were kept as compiled evidence and were not treated as new experiments.

2.3 Study selection

After title exclusion (154 records), retrieval failure (154 reports), and full-text exclusion (16 reports, including a tender-coconut-water file that was not the intended paper and a pork pet-food study excluded by protocol), 43 full texts were included: 34 from the bibliographic export that remained eligible after public-file retrieval, plus nine additional public papers from related search and targeted retrieval. Extraction used structured notes. Designs and endpoints are incommensurable; numbers were not meta-analysed across foods. Screening counts and fetch failures are recorded only here and in the run log.

3. Why the same setting is not the same kill

Because lethality depends on the physical state of the pack and on what the plate is allowed to count, ice, solute, injury, organic acids, and hold schedule are coefficients rather than footnotes. Primary challenges in this literature make that dependence measurable. Kinetic examples are summarised in Table 1.

Table 1. Coefficients that change the log reduction of a named hydrostatic cycle in included primary studies.

Study	Design	Matrix (n)	Cycle	Endpoint	Result
Bulut 2021 [9]	laboratory challenge	<i>E. coli</i> K12 in PBS and orange juice (juice n = 3)	400 MPa/9 min/4 °C, frozen vs unfrozen	log CFU/mL	unfrozen PBS 1.83 log; frozen −24 °C 5.63 log; frozen −80 °C 6.83 log
Torrents-Masoliver 2024 [10]	factorial challenge	<i>Listeria</i> , <i>Salmonella</i> , <i>E. coli</i> in apple vs banana–apple puree (two experiments)	300 MPa/2 min	log CFU/g, agar × time	banana–apple <i>Listeria</i> 0.3 ± 0.2 log immediately; apple 0.9- to 4.5-fold higher; Scott A 6.2 log when acid-exposed and sampled at 24 h
Nikparvar 2021 [11]	single-cell PI microscopy plus plates	<i>L. monocytogenes</i> Scott A in broth (318 cells; three count replicates)	400 MPa/8 min/8 °C	pore radius; log CFU/mL	7.79 ± 0.82 log plate reduction; mean pore radius 1.338 nm day 0 to 0.809 nm day 4
	Weibull kinetics	three <i>L. monocytogenes</i>		δ (min for first log)	control Scott A δ = 0.70 min; 2.8% lactate δ = 2.48 min; 10 min kill

Serra-Castelló 2021 [12]		strains on cooked ham	400 MPa, 0–10 min ± lactate		lower by 1.75–2.51 log with 2.8% lactate
González-Cantillo 2026 [13]	factorial burgers	lamb burgers ± lupine flour (n = 5 per cell)	600 MPa/4 min vs two 1 s pulses	mesophiles, colour	both schedules cut mesophiles from >7 to 3.7–4.5 log CFU/g; two short pulses whitened less
Scepankova 2022 [14]	spore challenge	<i>B. subtilis</i> in honey vs diluted honey	600 MPa/15 min ± 70–85 °C	log spores	undiluted honey (aW 0.57) unchanged at 5.65 log after 600 MPa/85 °C; PATP 600 MPa/15 min/75 °C took diluted aW 0.85–0.90 to the 1.30 log detection limit

Freezing before a modest hold changes kill because ice and the remaining unfrozen concentrate are not the same mechanical environment as a liquid pack. Unfrozen phosphate-buffered saline reached only 1.83 log of *E. coli* K12 after 400 MPa for 9 min at 4 °C, whereas the same cycle after freezing to –24 °C or –80 °C reached 5.63 and 6.83 log [9]. Freeze–thaw without pressure accounted for only about 1.5–1.7 log, so the extra lethality is the combination, not the freezer alone [9]. In orange juice, unfrozen 250 MPa treatments stayed at or below 0.42 log at pH 4.5 and 5.8, while frozen juice at 250 MPa for 15 min approached 4–5 log across those pH values [9]. Nucleic-acid leakage rose from about 28 ng/μL DNA in untreated cells to 182 ng/μL after freeze–thaw and 538 ng/μL after pressure on frozen cells, which is direct evidence of membrane and envelope failure rather than a plate-count artefact [9]. Scanning electron micrographs in that study are consistent with that leakage. The organism was K12, used as a relatively pressure-resistant surrogate, not Shiga-toxin-producing *E. coli*, so the result cannot be read as a validated STEC juice process [9]. Kinetic compilations of fruit and vegetable trials report the same direction: frozen carrot juice D-values for *E. coli* at 400 MPa were 2.12 min at –20 °C versus 5.32 min in unfrozen juice at 4 °C, and frozen orange juice at 250 MPa/15 min was compiled at 4.88 log versus 0.42 log without freezing [3]. Ice is therefore a coefficient. It is not a general licence to drop industrial pressure.

Water activity (aW) works in the opposite direction. In undiluted honey (aW 0.57), 600 MPa for 15 min at 85 °C left inoculated *Bacillus subtilis* endospores unchanged at 5.65 log, matching thermal-only 85 °C/15 min [14]. PATP at 600 MPa for 15 min at 75 °C on honey-water preparations reduced spores to the 1.30 log detection limit once aW reached 0.85–0.90 [14]. Ambient 600 MPa without that heat did not reduce total endospore load up to aW 0.90, and in the most diluted preparation it increased apparent spore counts, which the authors

interpret as activation of superdormant spores rather than growth [14]. Low aW and high soluble solids are compiled as baroprotectants in fruit-product kinetics as well: D-values for *Alicyclobacillus acidoterrestris* spores rose from 4.17 to 13.71 min as soluble solids went from 10 to 30 °Brix [3]. A 600 MPa label on a low-moisture or high-solids food is therefore not the 600 MPa label on a dilute juice.

What the laboratory chooses to count changes the apparent kill even when the vessel cycle is fixed. In hygienized fruit puree, 300 MPa for 2 min reduced a *L. monocytogenes* cocktail by only 0.3 ± 0.2 log in banana–apple puree when plates were read immediately, while the same organisms in apple puree (pH 3.32 versus a higher-pH banana mix) were inactivated 0.9- to 4.5-fold more [10]. Waiting 24 h before plating increased the apple-puree *Listeria* cocktail result from 1.8 ± 0.3 to 3.5 ± 0.2 log, and acid exposure for 24 h before pressure plus delayed plating took Scott A to 6.2 ± 0.1 log [10]. Single-strain *E. coli* on the same apple puree spanned 0.6 log (CECT5947) to 2.8 log (CTC1030) immediately after pressure [10]. Selective agar versus tryptic soy, and sampling time after decompression, therefore manufacture part of the log reduction. In already-pasteurised commercial orange juice re-inoculated with *E. coli* DSM682, 400 MPa for 2 min left about 2.5 log survivors while 400 MPa for 3 min took counts to not detected, and a prior ultrasound dose changed the starting number that pressure then had to finish [15]. Those two juice papers do not contradict each other once pressure, matrix, and whether the juice was raw or already pasteurised are named. They do show that “300–400 MPa on juice” is not a single microbial sentence.

Injury is not only an agar effect. After 400 MPa for 8 min at 8 °C in broth, *L. monocytogenes* Scott A fell 7.79 ± 0.82 log on plates, below the limit of quantification, while 318 treated cells remained propidium-iodide permeable [11]. Modelled mean pore radius declined from 1.338 nm on day 0 (95% Bonferroni interval 1.296–1.380) to 0.809 nm on day 4, and the least-damaged cluster’s fluorescence at 30 min fell seven-fold over those four days [11]. Counts stayed below quantification through day 4 and exceeded it only on day 7, which is consistent with a shrinking-pore repair trajectory rather than instantaneous complete kill [11]. The system is broth and one strain, so it cannot set a food specification, but it demonstrates the mechanism that VBNC-focused reviews warn about: culture “kill” can hide cells that later repair [6,16]. Compiled *Listeria* overviews report recovery to about 8 log in Queso Fresco by day 60 after an immediate below-detection result at 600 MPa/3 min, and delayed colony appearance after ≤ 550 MPa when plates are re-read days later [16]. A blank plate on day 0 is therefore not the same safety outcome as a blank plate after refrigerated storage.

Organic-acid hurdles that look like extra safety on a label can protect *Listeria* against pressure. On sliced cooked ham, 400 MPa inactivation of three strains was strain-specific in control product (Scott A $\delta = 0.70$ min with a resistant tail; CTC1011 $\delta = 5.98$ min with a

shoulder), and 2.8% potassium lactate raised those δ values and cut 10 min kill by 1.75–2.51 log relative to control ham [12]. The global model predicted that time to a second log reduction for Scott A rose from 2.10 min at 0% lactate to 11.22 min at 2.8% lactate [12]. Sodium diacetate at 0.1% moved the other way, reducing δ , and lactate plus diacetate together was not different from control [12]. Water activity of the hams did not explain the lactate effect. The authors describe piezo-protection; the measurement is a longer Weibull first-log time, not a photographed membrane event [12]. RTE-meat scoping evidence sits beside that primary curve: among 29 high-pressure *Listeria* papers in a 74-study map, reported log reductions ran from 0 to 7, and holds below 400 MPa were usually below 1 log [17]. Broth pores, ham lactate, and a scoping range of 0–7 log are not a disagreement about whether pressure works. They are measurements of different matrices, strains, and acids under one megapascal number.

Hold schedule is a coefficient even when total machine time is matched. In lupine-flour lamb burgers, one 600 MPa cycle of 4 min and two 1 s cycles at 600 MPa both cut day-1 mesophiles from more than 7 log CFU/g to 3.7–4.5 log and psychrophiles to about 1 log, with coliforms and *E. coli* below 1 log through day 14 [13]. The two short pulses whitened and oxidized less than the 4 min hold [13]. The trial is background flora, not a *Listeria* or *Salmonella* challenge, and the authors warn against reading it as process validation [13]. Pulse versus hold can therefore be similar for indigenous counts and different for colour in the same burger. Kinetic reviews that compile multi-pulse fruit-product data likewise treat pulse number as an interacting factor rather than a free substitute for dwell [3]. Taken together, these data suggest that ice, aW, agar, pores, lactate, and pulse are not noise around a true MPa effect. They are the effect. This section cannot show a single industrial cycle that absorbs those coefficients, and it cannot show freeze-then-pressure lethality for *Listeria* or STEC in juice, because the freeze primary used *E. coli* K12 [9,16].

4. Legal and five-log performance criteria versus what studies measured

A legal pasteurisation or five-log sentence is a performance claim. Included studies often measured something adjacent: indigenous aerobes, a mixed-model average, or a scoping range. Table 2 places those bars next to what was actually counted.

Table 2. Performance criteria and the microbial bars actually measured in included syntheses and food trials.

Study	Kind	Matrix	Stated or implied bar	What was measured	Result relative to that bar
Koutsoumanis 2022 [1]			legal milk pasteurisation	assembled inactivation	non-thermal HPP cannot match legal milk

	EFSA BIOHAZ opinion	ruminant milk; RTE cooked meat	(>10 log of named vegetative pathogens); RTE <i>Listeria</i> 2-log vs 5-log minima	and a <i>Listeria</i> model	pasteurisation; 600 MPa/6 min meets several milk PCs except <i>S. aureus</i> (about 6 log); RTE cooked meat ≥ 5 log <i>Listeria</i> at 600 MPa needs 4.7 min
Sykora 2026 [7]	PRISMA mixed-model, 89 articles	liquid bovine milk	FDA-style >5-log vegetative pathogens	log CFU/mL vs pressure	~450 MPa <i>L. monocytogenes</i> , ~500 MPa <i>S. enterica</i> , ~600 MPa <i>S. aureus</i> ; strain-averaged <i>E. coli</i> did not reach >5 log
Li 2021 [18]	industrial-style trial, sensory n = 101	whole Concord puree	~5-log indigenous APC/ Y&M, not named pathogens	APC, yeasts and moulds	both 600 MPa/3 min/5 °C and 63 °C/3 min ~5 log on day 1; HPP kept yeasts lower through month 4
Eran Nagar 2026 [19]	semi-industrial match	Red Globe juice, two ~30 L batches	matched ~3-log indigenous aerobes	PCA, yeasts, PPO	500 MPa/5 min $\approx$ 90 °C/1 min $\approx$ ohmic 120 °C/3 s for ~3 log aerobes; PPO left active after pressure
Machado 2023 [20]	two-batch cream trial	raw bovine cream	shelf-life vs 75 °C/15 s, not EU milk law	TAP, LAB, inoculated <i>E. coli</i> / <i>L. innocua</i>	600 MPa outlasted 75 °C/15 s for endogenous flora; 450 MPa/5 min did not
Koker 2023 [21]	factorial donkey milk	raw donkey milk, initial TABC 4.04 log	TABC reduction vs 75 °C	TABC, lysozyme, lactoferrin	500 MPa reduced TABC by 4.04 log (to the starting load); 200 MPa only 0.85 log
Guzel 2026 [17]	PRISMA-ScR, 74 studies	RTE meats, <i>L. monocytogenes</i>	no pooled effect	log reduction range	HPP in 29/74 papers; 0–7 log; <400 MPa usually <1 log

EFSA's opinion on non-thermal HPP answers a legal question, not a new challenge trial. Product temperature below 45 °C is in scope; homogenization, multipulse pressure, and HPTP above 45 °C are out of scope [1]. Non-thermal pressure does not inactivate spores. It cannot match legal thermal pasteurisation of milk, because that thermal standard is expected to give more than 10 log₁₀ of STEC, *L. monocytogenes*, *Salmonella*, *S. aureus*, and *Campylobacter* [1]. Example pressure–time combinations assembled for an 8-log performance criterion in milk put *S. aureus* at 600 MPa–8 min, 550 MPa–10 min, or 500 MPa–15 min, while STEC

and *Salmonella* sit near 600 MPa–5 min and *Campylobacter* near 600 MPa–1 min [1]. The most stringent industrial condition collected, 600 MPa/6 min, would meet those criteria except for *S. aureus*, for which it was judged to give about 6 log₁₀ [1]. Alkaline phosphatase is pressure-resistant and at best an overprocessing indicator; no suitable commercial enzyme marker was identified for 400–600 MPa / 1.5–6 min [1]. For RTE cooked meat, modelled *L. monocytogenes* minima at 90–95% certainty were ≥ 2 log at 600 MPa–2.3 min versus ≥ 5 log at 600 MPa–4.7 min [1]. Generic minima could not be set for soft cheese or smoked or gravad fish [1]. The opinion is therefore explicit that a 600 MPa label is not one legal outcome, and that milk is the sharpest mismatch.

The bovine-milk systematic review supplies the numerical scatter behind that mismatch without replacing the opinion. Across 89 articles (65 microbial, 24 protein, only four with both endpoints), mixed-model ANOVA found pressure to be the strongest lethality factor (*L. monocytogenes* $F = 171.12$, *S. aureus* $F = 87.73$, *E. coli* $F = 22.28$; all $P < 0.0001$), while time was not significant for *S. aureus* ($P = 0.06139$) [7]. Greater-than-five-log averages sat near 450, 500, and 600 MPa for *Listeria*, *Salmonella*, and *S. aureus* respectively; strain-averaged *E. coli* still missed the five-log bar [7]. At 301–450 MPa, *L. monocytogenes* reductions spanned 0.39–8.87 log across papers [7]. Pressurization without an adequate hold often failed five-log *Listeria* even above 551 MPa [7]. Milk fat can protect *E. coli*: a cited 600 MPa/15 min/20 °C cycle gave 1.6 log in 3.6% fat milk versus 3.0 log in skim [7]. Protein retention moves the other way. Average 50% denaturation of β -lactoglobulin occurred at 200–300 MPa, versus about 400 MPa for IgG, 600 MPa for alkaline phosphatase, and 800 MPa for lactoferrin [7]. One compiled cycle, 600 MPa/15 min/20 °C in mature skim milk, exceeded five log for *L. monocytogenes*, *Pseudomonas aeruginosa*, and *S. aureus* while leaving β -lactoglobulin at 20.1% and lactoferrin at 39.0% native [7]. Dairy narrative reviews of components and of recent milk-product work make the same qualitative point: 600 MPa/5 min is not interchangeable with 400 MPa/5 min for whey unfolding, casein micelle size, or sensory quality, and vegetative spores survive [22,23]. Those reviews do not generate a new paired 72 °C/15 s trial.

Primary dairy trials in the sample are not that missing paired milk challenge. Raw bovine cream at 600 MPa lasted longer in refrigerated storage than 75 °C/15 s for endogenous flora, with inoculated *E. coli* and *Listeria innocua* reduced by more than 6 log to the 1.00 log detection limit immediately after pressure, whereas 450 MPa/5 min behaved like untreated cream for total aerobic psychrotrophs [20]. Two batches differed in colour, and *E. coli* rebounded then collapsed during storage after 600 MPa/5 min [20]. Donkey milk started at only 4.04 log total aerobic bacteria, so 500 MPa produced a 4.04 log reduction to the floor of the starting load, while 200 MPa produced 0.85 log and 75 °C for 1–2 min produced 0.72–1.34 log [21]. Native lysozyme and lactoferrin were better retained at some moderate

pressures than after heat, which is a quality comparison, not a five-log *S. aureus* comparison [21]. Cream and donkey milk therefore show that 600 MPa can beat or match a 75 °C hold for native flora in those matrices. They do not re-run EFSA's milk conclusion on *S. aureus* in ruminant milk beside 72 °C/15 s.

Juice five-log language meets a similar gap. Whole Concord grape puree at 600 MPa/3 min/5 °C and mild heat at 63 °C/3 min both achieved about 5 log of indigenous aerobic plate count and yeasts and moulds from untreated day-1 levels of 6.33 and 6.31 log CFU/g [18]. Pressure kept yeasts and moulds below detection longer: month 4 yeasts remained below the limit after pressure while heat had 3.1 log, and by month 5 heat exceeded 6 log with visible mould [18]. Residual PPO and peroxidase after pressure were 75.2% and 80.7%, versus 22.6% and 10.2% after heat [18]. Consumer sensory scores (n = 101) and phenolic totals did not collapse after pressure [18]. Semi-industrial Red Globe juice was matched to a different bar: about 3 log of indigenous aerobes, taking 1.5×10^5 CFU/mL to $2.2\text{--}3.0 \times 10^2$ CFU/mL for pressure at 500 MPa/5 min, plate heat at 90 °C/1 min, and ohmic heating at 120 °C/3 s [19]. Yeasts fell 3–4 log under all three. PPO told another story: heat inactivated more than 90%, while pressure left 41 ± 8 versus 58 ± 13 min⁻¹ mL⁻¹ in fresh juice, a non-significant ~30% reduction [19]. Colour after pressure was not different from fresh; thermal processes produced ΔE about 6.4–6.5 [19]. Concord 600 MPa/3 min is therefore not Red Globe 500 MPa/5 min, and neither is a named-pathogen five-log juice validation. Infant-food puree at 300 MPa/2 min, discussed in Section 3, is further from that industrial bar [10].

RTE *Listeria* evidence is a range, not a legal minimum copied onto every meat. The scoping review did not pool a single effect size; 0–7 log across 300–600 MPa (occasionally 800 MPa) is the reported span, with holds generally under 15 min at 4–25 °C [17]. Compiled overviews add matrix extremes: 500 MPa/10 min/25 °C below detection in UHT milk and mozzarella but only 1.50 log in smoked salmon; dry-cured ham at aW 0.88 only 0.80–1.50 log even at 450–600 MPa; a six-strain beef cocktail 3.90 log at 500 MPa/2 min/4 °C versus ≥ 6.50 log at 7 min [16]. Seaweed-fortified frankfurters at 600 MPa/6 min extended time to 6.49 log from under 15 days to about 45 days without an inoculated-pathogen arm [24]. Precooked crab lump and roe with a chitooligosaccharide–catechin conjugate reached a different pressure story: 100 MPa left about 5–6 log aerobic plate count, 300 MPa cleared several groups but not all aerobes, and 500 MPa for 1.5 or 3 min brought tested groups to zero on the authors' table, while methods text listed 100 and 300 MPa rather than 500 MPa [25]. That labelling disagreement is a reason not to treat the crab row as a clean 500 MPa validation. Taken together, these data suggest that legal and five-log sentences have to be earned in the food and organism of interest. This section cannot show a retrieved primary HPP-versus-72 °C/15 s *S. aureus* milk trial, and it cannot show a public industrial 600 MPa juice challenge against *Salmonella*, STEC, or *Listeria*.

5. Spores, heat-assisted pressure, and cold-chain hurdles

Ambient hydrostatic pressure and heat-assisted pressure share a megapascal number and do not share a spore outcome. Table 3 records that split.

Table 3. Spore and hurdle outcomes under ambient pressure versus heat-assisted or combined cycles.

Study	Design	System	Cycle	Endpoint	Result
Knoerzer 2025 [4]	engineering review	water-rich and fatty foods	HPTP, adiabatic heating	temperature rise	~3 °C per 100 MPa in water-rich foods (~18 °C at 600 MPa); fats heat more
Scepankova 2022 [14]	inoculated spores	honey vs diluted honey	600 MPa/ 15 min ± 70–85 °C	<i>B. subtilis</i> TEL	no kill in undiluted honey; ≥3.7 log to LOD at aW 0.85–0.90 with 75 °C
Heydenreich 2024 [26]	germination–inactivation, n ≥ 3	<i>B. subtilis</i> and <i>B. amyloliquefaciens</i> in ACES buffer	550 MPa/ 60 °C/2.5 min ± nisin	log10 N/N0, flow cytometry	<i>B. subtilis</i> vHP –2.1 log before heat, –5.7 after heat; +2,500 IU/mL nisin –5.3 before heat; <i>B. amyloliquefaciens</i> ~2 log on the best <i>B. subtilis</i> combination
López 2026 [27]	storage ± 24 h abuse	pumpkin–carrot cream, native flora, n = 3	600 MPa/ 8 min/55 °C ± enterocin AS-48	TSA log CFU/g	abuse took controls to 7.73 log by day 7; 55 °C pressure kept counts <1 log through 30 days with or without abuse
Le 2023 [28]	3×3 juice storage	celery ± lime or apple, “at least two” replicates	400–600 MPa/5 min	TAC, PPO/ POD	TAC <2 log CFU/mL in all three juices; enzymes and colour did not follow that kill
Nilsuwan 2024 [29]	hurdle, sensory n = 50	squid mantle ± mint extract	400 MPa/ 5 min	TVC, 16S	400 MPa without mint did not hold TVC to 6 log past day 6; mint plus pressure extended that window
Puangwerakul 2024 [30]	comparative preservation	plant-based egg analogue, n = 3	500 MPa/ 5 min vs gamma/ EBI	<i>C. perfringens</i> , total bacteria	all three selected doses zeroed day-0 counts; accelerated 40–50 °C storage is not retail

Adiabatic compression heating is the reason HPTP can reach spore lethality that ambient holds miss. The engineering review of that class treats the temperature rise as ideally uniform work of compression, warns that scale-up and packaging change the heat that is actually kept, and places spore inactivation in the heat-assisted window rather than in chilled 600 MPa juice cycles [4]. Plant-food kinetic compilations agree on the spore limit: vegetative cells at 400–600 MPa, spores often surviving below 600 MPa at ambient temperature, and *B. subtilis* spores reported to survive up to 1200 MPa at ambient temperature in cited work [2,3]. *A. acidoterrestris* spores in orange juice were compiled at less than 1 log at 600 MPa/45 °C/10 min versus 2.7 log at 65 °C; *Neosartorya fischeri* ascospores in apple juice showed no reduction at 0.1 MPa/75 °C versus 4.8 log at 600 MPa/75 °C in one tabled comparison [3]. *Bacillus coagulans* spores in tomato pulp rose from 2.5 log at 300 MPa/60 °C/20 min to 6.5 log at 600 MPa/60 °C/20 min [3]. Those numbers come from heterogeneous primaries inside a narrative review. They still show that temperature during the hold is not optional for spores.

Buffer germination studies explain one proposed pathway without converting it into a food sterilization claim. Very high pressure at 550 MPa, 60 °C, 2.5 min germinated a large fraction of *B. subtilis* spores in ACES buffer (about –5.7 log after a scoring heat that selects ungerminated spores) while only partly inactivating them without nisin (–2.1 log before that heat) [26]. Adding 2,500 IU/mL nisin moved before-heat kill to –5.3 log while after-heat stay near –5.7, so nisin inactivated germinated spores rather than raising the germinated fraction under those conditions [26]. Two very-high-pressure cycles with nutrient and nisin and a 37 °C hold reached about 8 log reduction of culturable dormant *B. subtilis*, still with an estimated tail of fewer than 40 of 10⁹ spores/mL ungerminated [26]. The same named combination gave only about 2 log in *B. amyloliquefaciens* [26]. Moderate pressure at 150 MPa/37 °C/5 min is discussed as a germinant-receptor route; very high pressure as dipicolinic-acid release independent of those receptors [26]. The matrix is buffer pH 7.0, the nisin dose is above typical food use, and *Clostridium* was not tested. Combination mini-reviews of plant foods make the practical translation: single room-temperature pressure is limited for spores and some Gram-positives, so thermal, ultrasound, ultraviolet, antimicrobial, pulsed-electric-field, or carbon-dioxide pairings are proposed as patterns, not as interchangeable unit operations [8].

When heat is allowed in a real food, the cycle must be labelled as heat-assisted. Pumpkin–carrot vegetable cream treated at 600 MPa for 8 min with a 55 °C hold kept native aerobic mesophiles below 1 log CFU/g for 30 days at 4 °C even after a 24 h room-temperature abuse on day 2, whereas abused untreated cream rose from about 2.5 log to 6.47 log by day 3 and 7.73 log by day 7 [27]. Enterocin AS-48 at 50 µg/g also blocked that bloom. Synergy of bacteriocin plus pressure could not be shown because starting counts were already near 2.5 log [27]. PMA-16S of untreated cream shifted toward *Bacillus*-dominated Bacillota without

abuse and retained more *Acinetobacter*-associated Pseudomonadota after abuse; many pressurized samples had too little PMA-surviving DNA to sequence [27]. Authors chose 55 °C because cold pressure often fails on spores and pressure-tolerant enzymes [27]. That 600 MPa/55 °C cream cycle is therefore not the 600 MPa chilled juice cycle in Section 4.

Hurdles that stay cold still change what a named hold means for shelf-life flora, without becoming spore sterilization. Celery juices acidified with lime (pH 3.30) or apple (pH 4.50), or left near pH 6.02, all fell below 2 log total aerobic count after 400–600 MPa for 5 min, but PPO, peroxidase, colour, and phenolics did not move in lockstep with that microbial floor over 8 weeks at 4 °C [28]. Replication was described as “at least two,” which is thinner than a conventional triplicate [28]. Squid mantle at 400 MPa for 5 min reduced planktonic *Pseudomonas aeruginosa* and *Shewanella* from about 7 log to near 2 log, yet without mint extract total viable counts did not stay below 6 log past day 6; mint extract plus pressure extended sensory and count keeping, with a 50-person panel [29]. Sous-vide Atlantic cod plus 300–600 MPa kept tryptic-soy counts down through mid-storage, with 500 MPa/8 min below 1.30 log through day 45 in one trial, but counts returned by day 60–75 and the authors explicitly call for a *C. botulinum* endospore challenge that they did not run [31]. Hyperbaric storage at 50–150 MPa in honey did not kill *B. subtilis* spores; at 150 MPa it prevented outgrowth that occurred at atmospheric pressure [14]. Those are cold-chain or hurdle results.

One plant-based egg analogue zeroed day-0 total bacteria, yeasts, and *Clostridium perfringens* after 500 MPa/5 min, matching selected gamma and electron-beam doses on that day-0 plate [30]. Accelerated storage at 40–50 °C is not retail refrigeration, and *C. perfringens* in a rice-protein analogue is not proteolytic *C. botulinum* in a low-acid canned food [30]. Aquatic-food reviews of ultra-high pressure similarly treat vegetative spoilage and parasite reduction as pressure–time–temperature–matrix dependent and leave spores standing at room-temperature industrial settings [32]. Taken together, these data suggest that spore control, when it appears at all, is heat-assisted pressure, raised aW, or a different organism in a different analogue. This section cannot show proteolytic *C. botulinum* kinetics under true non-thermal HPP in a food.

6. Quality endpoints that do not track plate counts

Microbial success is not a quality specification. Enzymes, colour, texture, proteins, aroma, and phenolics can move against, or independently of, the plate. Table 4 collects comparable quality contrasts.

Table 4. Quality endpoints that diverge from plate-count success under named hydrostatic cycles.

Study	Design	Food (n)	Cycle		Result
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				Quality endpoint	
Eran Nagar 2026 [19]	semi-industrial match	Red Globe juice (PPO n = 9)	500 MPa/5 min vs 90 °C/1 min	PPO; ΔE	pressure ~30% PPO loss, colour like fresh; heat >90% PPO loss, ΔE ~6.4
Li 2021 [18]	parallel preservation	Concord puree (sensory n = 101)	600 MPa/3 min vs 63 °C/3 min	PPO/POD; consistency	pressure left 75–81% oxidase activity; heat gelled by month 4
Tsikrika 2021 [33]	commercial HPP ± ultrasound	Maris Piper and Rooster potatoes (n = 6)	600 MPa/3 min	ΔE, phenolics	day-14 Rooster ΔE 8.06 untreated vs 21.23 HPP vs 18.03 HPP/ultrasound
Jönsson 2023 [34]	laboratory HPP	<i>Saccharina latissima</i> (texture n = 6)	200–600 MPa/180 s	hardness	hardness –67.7% at 200 MPa to –87.7% at 600 MPa; colour retained
Wang 2025 [35]	UHP vs retort	pre-cooked mackerel balls (sensory n = 8)	250/350 MPa/10 min vs 121 °C	hardness; counts	both pressures <1 log CFU/g day 0 vs retort 1.74–1.77; hardness differed by cycle
Koker 2023 [21]	HHP vs 75 °C	donkey milk (two treatment replicates)	200–500 MPa	lysozyme, lactoferrin	500 MPa hit the TABC floor; native proteins better retained at some moderate pressures than after heat
Rossi Ribeiro 2026 [36]	factorial pulp	pumpkin pulp ± peel preservative (triplicate)	600 MPa/4 min vs heat	texture, β-carotene, 45-day flora	formulated HPP pulps undetectable microbes 45 days; thermal pulp much firmer and lost detectable β-carotene by day 45

PPO is the clearest split. When Red Globe juice processes were matched to the same ~3-log aerobic reduction, heat destroyed the enzyme and the colour, and pressure kept both nearer to fresh juice [19]. Concord puree showed the same oxidase residual after a more severe 600 MPa/3 min cycle that had already matched mild heat for indigenous five-log counts [18]. Celery juices lost aerobic counts at 400–600 MPa/5 min while PPO and peroxidase stayed matrix- and pH-dependent [28]. Potatoes at commercial 600 MPa/3 min, with or without a laboratory ultrasound step, shifted *a*, *b*, *L**, and ΔE versus untreated packs, with Rooster ΔE at day 14 rising from 8.06 untreated to 21.23 after pressure [33]. Enterobacteriaceae outcomes and chlorogenic acid also differed when ultrasound followed pressure, so 600 MPa/3 min is not 600 MPa/3 min plus ultrasound for either colour or those phenolics [33]. Sugarcane-juice compilations list 300 MPa/2–5 min with PPO residual 92.77 and a 6.32 log plate-count drop,

versus 523 MPa/50 °C/11 min with 62% PPO inactivation and a 5-log microbial reduction: the hotter, longer, higher-pressure row is already HPTP-like [37].

Texture can collapse while colour holds, or the reverse. Cultivated *Saccharina latissima* lost 67.7% hardness at 200 MPa/180 s and 87.7% at 600 MPa, with compression distance falling in parallel, while colour was retained and indigenous microbial load sat near enumeration limits [34]. Microbial n was two bags per pressure in the refrigerated arm, so the kill claim is weak and the texture claim is the one the design can carry [34]. Pre-cooked mackerel balls at 250 or 350 MPa for 10 min went from 2.18 log CFU/g untreated on day 0 to below 1 log, while 121 °C retort left 1.74–1.77 log; hardness in newtons and malondialdehyde still separated those cycles, and eight panellists scored sensory differences that plate counts alone would not have ranked [35]. Pumpkin pulp at 600 MPa/4 min with an acid/starch mix and optional peel extract stayed below detection for mesophiles, Enterobacteriaceae, and yeasts for 45 days at 5 °C, whereas a previous same-pressure run on ingredient-free pulp had a 15-day shelf-life; thermal pulp was up to several-fold firmer and lost detectable β -carotene by day 45 [36]. The same 600 MPa/4 min is therefore not one shelf-life or one texture without the mix.

Proteins unfold on a non-covalent path that is not a plate count. A 2014–2024 protein review concludes that high hydrostatic pressure disrupts non-covalent bonds, exposes hydrophobic interiors, often raises surface hydrophobicity and antioxidant activity, and often lowers allergenicity in plant and seafood proteins, with the same megapascals failing that pattern in some nuts [38]. Buckwheat allergenicity was compiled as 72.2% reduced at 400 MPa/20 min, while almond extracts at 400–580 MPa/3 min showed increased immunoreactivity [38]. Milk-component reviews describe whey unfolding, fat-globule disruption, and colour or turbidity shifts that depend on whether 400 or 600 MPa is used [22]. Donkey-milk lysozyme and lactoferrin in the primary factorial trial are a small-n illustration of that protein-versus-count split [21]. None of those protein outcomes licenses a microbial pasteurisation claim, and none should be averaged with juice PPO.

Aroma and phenolics are two further quality jobs that a named cycle does not freeze. Lipid-derived aldehydes, alcohols, ketones, esters, lactones, and terpenes change under 100–1000 MPa relative to untreated or heated foods in a compound-class review; ranges such as 3–261% mix many products and gas-chromatographic units, so the supported statement is that pressure is not aroma-neutral, not that a particular percentage applies to every food [39]. Phenolics split by job: mild pressure (often 15–100 MPa, 10–20 min) is discussed as an abiotic elicitor of biosynthesis via phenylalanine ammonia-lyase and reactive oxygen species, whereas higher pressure improves extraction after cell disruption [40]. Several biosynthesis papers lack those assays, which is why the mechanism is proposed rather than demonstrated

in every cited row [40]. Fermented jujube juice pretreated at 500 MPa/10 min with or without ultrasound, versus 85 °C/15 min, changed browning, ascorbic acid, colloidal stability, and volatile profile during lactic fermentation; that trial is a quality pretreatment, not a pathogen challenge [41].

Fish fillets and aquatic reviews close the quality map without turning it into a microbial meta-analysis. Salmon and plaice at 500 MPa/2 min/4 °C kept treated fillets microbiologically acceptable to day 12 versus day 8 for vacuum-packed controls, with histamine remaining low and a five-person laboratory panel (COVID-restricted) scoring sulfur off-odours lower in treated fish [42]. Aquatic ultra-high-pressure reviews of 2012–2022 compile vegetative inactivation and quality change in fish, shrimp, and oysters and again leave spores at room-temperature industrial settings [32]. Taken together, these data suggest that oxidase residual, colour ΔE , texture collapse, whey unfolding, aroma shifts, and phenolic extraction or biosynthesis are separate endpoints from log CFU. This section cannot show energy use, package integrity, or consumer willingness-to-pay, which were not measured.

7. Neighbouring operations only to refuse false equivalence

Ultrasound, irradiation, ohmic heating, and homogenization appear in this literature beside pressure. They are included here so that a shared food name is not mistaken for a shared kill mechanism.

Homogenization is the sharpest false friend because it shares the words high pressure. Narrative distinction of high-speed blending versus valve homogenization places HPH in the shear-through-orifice class, often 20–100 MPa and sometimes higher, for liquids and emulsions rather than for isostatic packs [5]. The primary apple-juice homogenization trial retrieved to fill that gap used 100, 150, and 200 MPa with valve cooling so that product temperature stayed at or below 35 °C [43]. Fresh juice total mesophilic count was 5.36 log CFU/mL; homogenization cut that count by 0.6–1.4 log, with the 1.4 log maximum at 200 MPa, and extra passes at 150 MPa did not significantly change mesophiles, yeasts, or moulds [43]. After 7 days at 4 °C, yeasts on 150 MPa juice rose 1.2 log; by day 21 total count exceeded the fresh-juice start [43]. PPO fell 21.5% and peroxidase 16.4% at 200 MPa. Vitamin C (dehydroascorbic acid) retained 63.8% at 100 MPa and 38.2% at 200 MPa; total phenolics rose about 20% at 150 MPa and fell slightly at 200 MPa [43]. Particle-size $D_v(50)$ dropped from 308 μm to 15.8 μm already at 100 MPa [43]. That is a colloid and modest native-flora result in a cooled liquid. It is not hydrostatic 600 MPa pasteurisation, and it is not a five-log pathogen claim.

Ultrasound plus pressure is a sequence, not a synonym. In already-pasteurised orange juice, power ultrasound alone took *E. coli* DSM682 from 5.1 log down by 1.67–1.71 log after 22.1

min, and 400 MPa finished or failed depending on two versus three minutes [15]. Potatoes showed ultrasound after 600 MPa changing colour and Enterobacteriaceae relative to pressure alone [33]. Jujube fermentation used ultrasound-assisted 500 MPa as a pretreatment relative to 85 °C [41]. Sugarcane-juice and plant-food combination reviews catalogue those pairings beside pulsed electric fields, ultraviolet, ozone, and membranes [8,37]. Equal-time chapters for each of those technologies would hide the hydrostatic question this review is asking. Irradiation versus pressure in the plant-based egg analogue zeroed the same day-0 plates and then diverged in accelerated storage, which is a reminder that matched day-0 zeros are not matched processes [30]. Ohmic and plate heat in Red Globe juice were microbial matches to 500 MPa/5 min and enzyme opposites [19]. None of those neighbours should be pooled with hydrostatic log reductions.

The refusal is practical. A processor who copies 200 MPa from a homogenizer manual onto an isostatic vessel, or 600 MPa from a chilled juice onto a honey spore problem, or a 1 s double pulse from a burger colour trial onto a *Listeria* ham specification, is copying a number that the included measurements do not support. Neighbouring operations can be useful hurdles. They are not the same machine.

8. Discussion

The included evidence jointly supports a modest, design-graded claim. Ambient industrial HPP in high-moisture foods reduces many vegetative bacteria and yeasts in the 400–600 MPa, few-minute window used in the EFSA opinion and in primary storage trials of puree, cream, sausages, fish, and burgers [1,13,18,20,24,42]. That claim is not a pooled log reduction and is not a sterility claim. Bacterial spores remain outside ambient holds [1,3,4,14]. Quality enzymes and structures often remain active or collapse independently of the plate [18,19,34,38]. Legal five-log juice language and legal milk pasteurisation are performance criteria that several included trials never measured directly [1,7,18,19].

The pattern exists because the causal chain is long. Named cycle, then membrane and protein physics (demonstrated as pores and nucleic-acid leak in broth and frozen juice; proposed as Ice I–III damage in kinetic reviews), then a plate count that depends on agar, time, and VBNC, then a quality assay on a different macromolecule, then a legal sentence that names an organism and a log bar the experiment may not have used [3,6,9,11]. Heterogeneity is therefore information. Banana–apple puree at 300 MPa/2 min is not already-pasteurised orange juice at 400 MPa/3 min [10,15]. Donkey milk that starts at 4 log cannot demonstrate a 5-log *S. aureus* bar [21]. Cream that outlasts 75 °C/15 s is not ruminant milk under EU thermal law [1,20]. Lactate on ham lengthens *Listeria* survival at 400 MPa while diacetate shortens it [12]. Two 1 s pulses at 600 MPa match a 4 min hold for burger mesophiles and not

for colour [13]. Frozen K12 juice is not a STEC validation [9]. Homogenized apple juice at 200 MPa is not hydrostatic 200 MPa [5,43].

Mechanism papers, outcome papers, and reviews fit if they are ranked. Nikparvar's pores and Bulut's leakage are demonstrated membrane events [9,11]. Serra-Castelló's lactate effect is a kinetic demonstration whose molecular piezo-protection remains proposed [12]. Wu and Wiśniewski compile VBNC and recovery as a risk that primary plates can miss [6,16]. Zhang, Houška, and Knoerzer compile spore limits and adiabatic heat [2,3,4]. Sykora and Koutsoumanis are the milk hinge, one mixed-model and one legal opinion, and they agree that 600 MPa is not automatically enough for *S. aureus* relative to a thermal pasteurisation bar [1,7]. Guzel refuses to pool RTE *Listeria* [17]. Quality reviews (Braspaiboon, Lomelí-Martín, Navarro-Baez, Serna-Hernandez, Agriopoulou, Peng) should be read as maps of endpoints, not as new n [22,23,32,38,39,40]. Heydenreich and López add spore germination and a 55 °C cream hurdle without filling the *C. botulinum* food hole [26,27]. Szczepańska fills the homogenization hole as a primary liquid trial [43].

What cannot currently be concluded is as important as what can. Non-thermal HPP has not been shown, in a retrieved primary paired trial, to match 72 °C/15 s against *S. aureus* in ruminant milk; EFSA's synthesis says it does not [1,7]. A public industrial 600 MPa juice study against named five-log pathogens is not in the included full texts [18,19]. Proteolytic *C. botulinum* under non-thermal pressure in a food is not in those texts [14,26,31]. Freeze-then-pressure for *Listeria* or STEC in juice is not in those texts [9,16]. Energy, packaging integrity, and consumer willingness-to-pay were not measured. Equipment cost figures that appear in some reviews are industry statements, not primary data, and they are not used here as findings. Numbers that share a log unit cannot be pooled across puree, milk, ham, honey, and buffer because organism, aW, fat, heat, and assay differ, as Table 1 through Table 4 show.

The practical implication is narrower than a slogan about nonthermal processing. Chilled commercial HPP can be a useful vegetative and shelf-life tool in high-moisture foods when the cycle is validated in that food with the organism and recovery method that matter. It is not a drop-in legal pasteurisation of milk, not a generic juice five-log stamp at every pressure, and not a spore sterilization unless heat, aW, or a different process class is added. The measurement that would close the sharpest hole is a public paired trial of industrial HPP versus 72 °C/15 s on *S. aureus* (and the other milk pathogens in the legal list) in liquid bovine milk, with come-up rate and fat reported, beside a juice challenge at industrial 600 MPa against the pertinent five-log pathogen with delayed and selective plating. A food-matrix *C. botulinum* spore curve under truly non-thermal holds would close the other hole that buffer *Bacillus* work cannot close.

9. Conclusions

A labelled hydrostatic cycle is a pressure, a time, and a temperature history, not a pasteurisation certificate. Vegetative kill in high-moisture foods at 400–600 MPa is supported by an opinion, a milk systematic review, and multiple primary storage and challenge trials, always with matrix and assay as coefficients. Spores, several oxidases, and legal milk and juice performance criteria do not travel with that vegetative success. Homogenization, ultrasound, and irradiation can sit beside pressure in the same plant; they do not inherit its log claims. The next useful experiments are paired milk HPP versus 72 °C/15 s on *S. aureus*, industrial 600 MPa juice challenges against named pathogens with injury-aware plating, and proteolytic *C. botulinum* curves in food under holds that either exclude or fully report adiabatic heat. Those are missing measurements, not a general request for more papers.

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